

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
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Driven by a problem solving mindset and fueled by curiosity, I always aspire to develop solutions for the real-world, impactful problems.

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering Budapest, Hungary

Supervisor: [Márton Vaitkus](#)

Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics Ulaanbaatar, Mongolia

GPA: 3.2/4.0 (**rank 3/34**)

Supervisor: [Galtbayar Artbazar](#)

Thesis: [Evaluating land degradation using image processing methods](#)

EXPERIENCE

Center of Mathematics Application, NUM Oct, 2023 – June, 2025

Undergraduate Research Intern

- **Technical Documentation:** Authored and maintained comprehensive lab documentation and user guides, enabling onboarding for new students and establishing clear protocols for research operations. Served as Lab Lead for a team of 6-8 students; collaborated to distill and present complex mathematical concepts during weekly seminars.
- **Data Management:** Architected a unified, well-documented data folder structure for research backups, directly improving data retrieval efficiency and creating written guidelines for future data integration.
- **Programming:** Engineered a custom [Python program](#) for processing drone imagery which is used to obtain the necessary vegetation index from drone imagery data. Worked in the Linux Environment with version control to ensure reproducibility.
- **Contributions:** Incorporated an open-source Docker containerized application (OpenDroneMap) for drone orthomosaic processing instead of proprietary software.

PROJECTS

[aiSim Data Integration for 3D Gaussian Splatting](#)

- Implemented a JSON parser in Python to analyze and convert complex sensor measurements from aiMotive's [aiSim](#) simulation software into a 3D scene representation ([nerfstudio](#)) format.

[Comparative EDA: Housing Affordability in Hungary vs Mongolia](#)

- Performed end-to-end exploratory data analysis (EDA) on public datasets from the [Hungarian Central Statistical Office \(KSH\)](#) and [National Statistics Office of Mongolia](#) to determine: “How many m^2 of a new apartment could a monthly average net wage buy, by region and year?”

SKILLS

Languages: English (C1 [IELTS 8.0](#)), Mongolian (native), Hungarian (A2)
Programming: Python, Bash, Julia, C/C++, R, MATLAB
Tools: Git, GitHub, Linux, Docker, Kubernetes, L^AT_EX